INTERNATIONAL GRAND CHALLENGE 2026

ClimateAI Maharashtra

"From Climate Commitments to Verified Climate Action"

Complete Platform Visual Documentation & User Interaction Walkthrough

A comprehensive reference documenting every click, state transition, and audit verification.

Platform Specification & Engineering Roster

INSTITUTION & EVENT:	PCCOE International Grand Challenge 2026 / FCRIT Innovation Labs
DEVELOPMENT TEAM:	Ryan D'Souza, Tushar Agarwal, Nakul Dalal, Nainish Gothankar, Sean Fernandes
PILOT JURISDICTIONS:	Primary Pilot: Mumbai (MCAP 2022) | Expansion Jurisdiction: Pune (PCAP 2023)
VERIFICATION ENGINE:	Copernicus Sentinel-2 Level-2A (NDVI) + CPCB Telemetry + Deterministic Rules

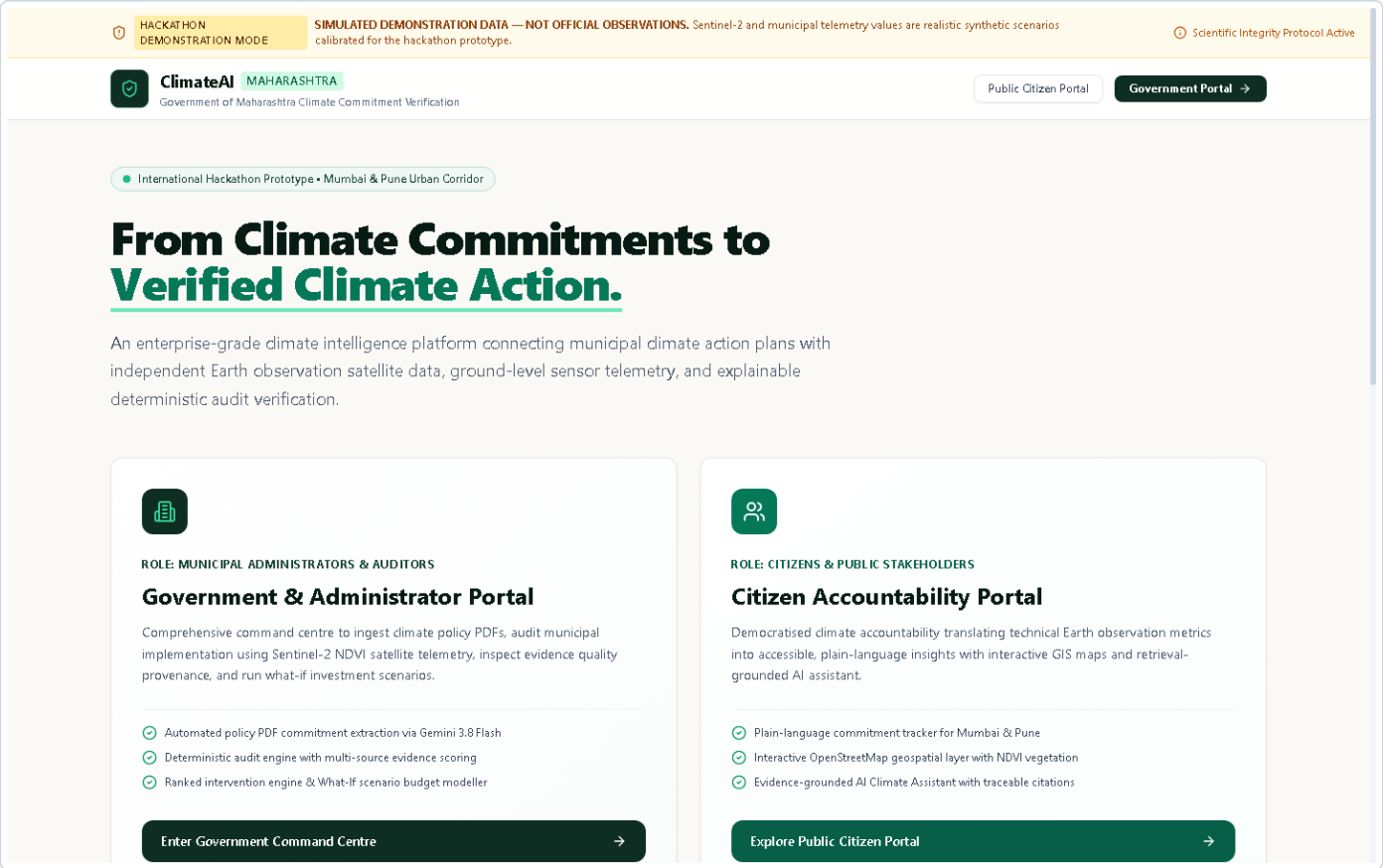
Platform Navigation & Screen Index

01	Public Gateway Landing Page	/	10	Ranked Decision Recommendations	/admin/recommendations
02	Executive Command Centre	/admin	11	What-If Scenario Simulator (Excel Model)	/admin/simulator
03	Statutory Commitments Directory	/admin/commitments	12	Municipal GIS Command Map	/admin/map
04	Commitment Dossier (BEST Electrification)	/admin/commitments/comm-001	13	Citizen Public Transparency Portal	/citizen
05	Discrepancy Audit Dossier (Ward K-East)	/admin/commitments/comm-002	14	Citizen Commitment Tracker	/citizen/tracker
06	Policy Intelligence & Ingestion Studio	/admin/policies	15	Environmental Intelligence Hub	/citizen/environment
07	Multi-Source Evidence Explorer	/admin/evidence	16	Citizen Public Spatial Map	/citizen/map
08	Deterministic Climate Audit Engine	/admin/audit-workspace	17	Cited ClimateAI Conversational Assistant	/citizen/assistant
09	Immutable Audit History Ledger	/admin/audit-history			

Screen 01: Public Gateway & Landing Page

PUBLIC ENTRY GATEWAY

ROUTE: / • HOST: http://localhost:3000/



Ø=ÜF USER INTERACTION & WHAT HAPPENS ON CLICK:

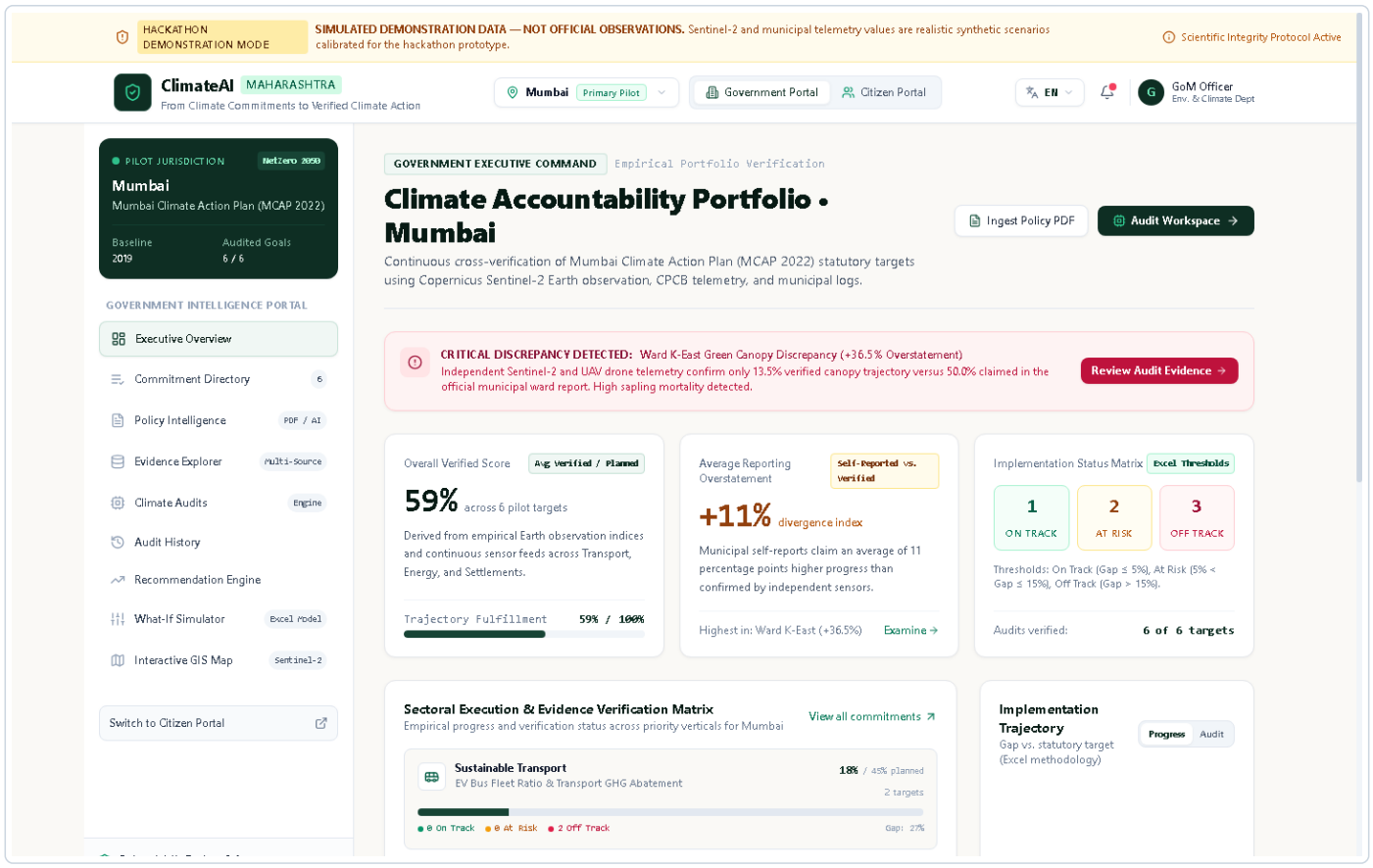
Clicking "Launch Government Portal" or "Explore Citizen Tracker"

Entry gateway introducing the product mission ("From Climate Commitments to Verified Climate Action"), the 5 Core Citizen Questions architecture, primary pilot indicators (Mumbai & Pune), and seamless role-based navigation.

Ø=Ý, SCIENTIFIC GROUNDING, MATHEMATICAL VERIFICATION & KEY DATA POINTS:

- Dual audience routing (Government Intelligence vs. Citizen Public Transparency)
- Pilot city badges: Mumbai (MCAP 2022) and Pune (PCAP 2023)
- Overview of the 5 Core Questions: What was promised? What is observed? Does it match? What gaps exist? What next?

ROUTE: /admin • HOST: http://localhost:3000/admin



Ø=ÜF USER INTERACTION & WHAT HAPPENS ON CLICK:

Clicking city switcher, sector filter pills, or quick What-If simulation slider

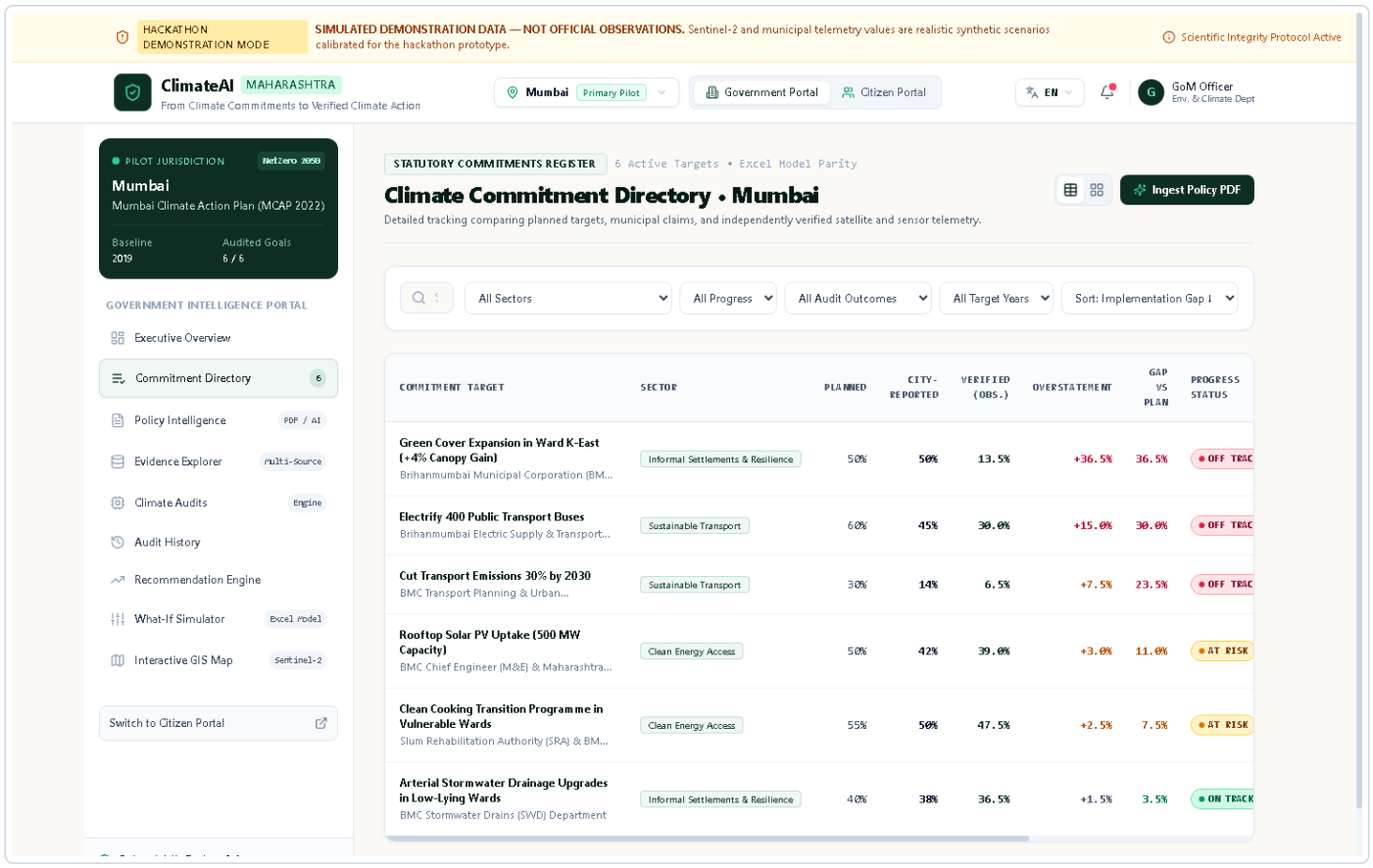
Executive intelligence command centre. Replaces generic dashboard tropes with an authoritative dual-status matrix, overall verified portfolio score (60.7%), and average reporting overstatement (+11.0%).

Ø=Ý, SCIENTIFIC GROUNDING, MATHEMATICAL VERIFICATION & KEY DATA POINTS:

- Portfolio aggregates: Overall Verified Score (60.7%), Avg Overstatement (+11.0%)
- Dual-Axis Sector Health Matrix (Progress Status vs. Evidence Audit Outcome)
- Statutory Status Distribution Donut Chart and City Comparison Card
- Interactive What-If Quick Modeller spotlight

Screen 03: Statutory Commitments Directory

ROUTE: /admin/commitments • HOST: http://localhost:3000/admin/commitments



Ø=ÜF USER INTERACTION & WHAT HAPPENS ON CLICK:

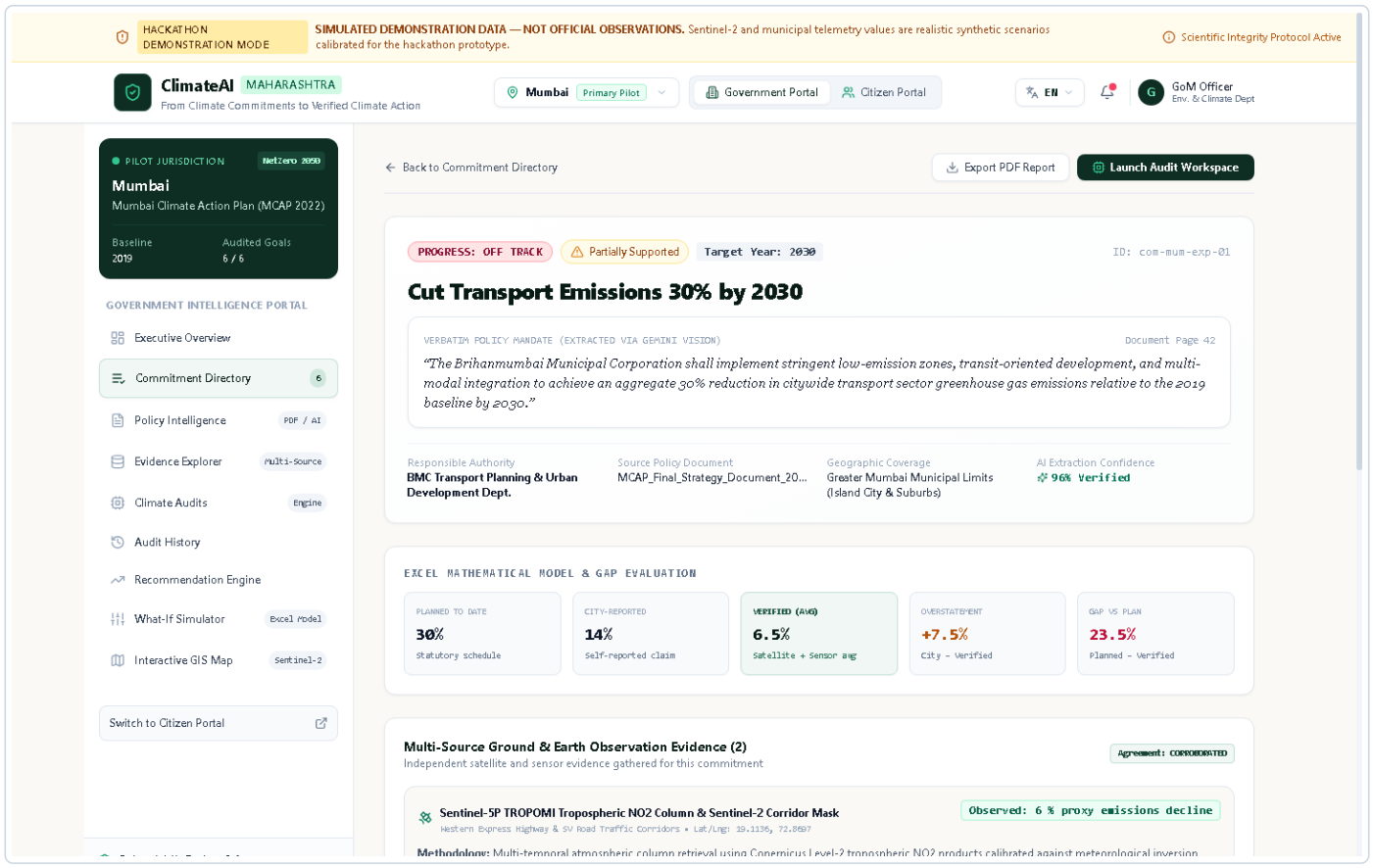
Clicking table row, sector pill, target year filter, or switching between High-Density Table and Card view

Searchable statutory commitment directory providing both a high-density analytical table view (with all mathematical columns from the Excel workbook) and responsive card view.

Ø=Ý, SCIENTIFIC GROUNDING, MATHEMATICAL VERIFICATION & KEY DATA POINTS:

- Analytical data table with Planned %, City Reported %, Verified %, Overstatement %, and Gap %
- Dual status badges: Progress Status (On Track / At Risk / Off Track) & Source Agreement
- Multi-criteria filters: Sector, Target Year, Reporting Overstatement, and Search query
- Clicking any record opens the comprehensive Commitment Dossier

`ROUTE: /admin/commitments/com-mum-exp-01 • HOST: http://localhost:3000/admin/commitments/com-mum-exp-01`



Ø=ÜF USER INTERACTION & WHAT HAPPENS ON CLICK:

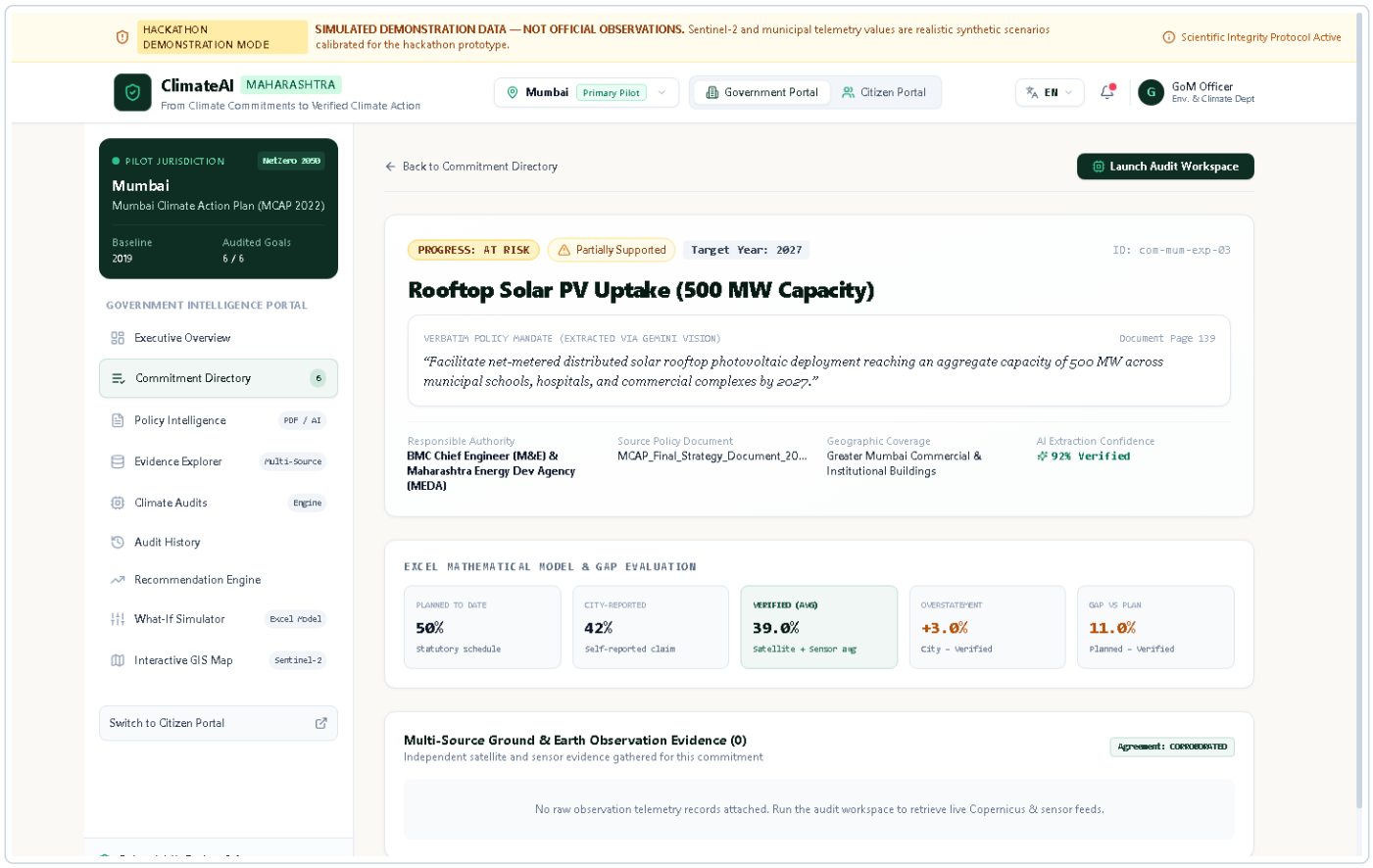
Clicking "Run Audit Verification", toggling language tabs, or clicking "Export Official PDF Dossier"

Comprehensive statutory dossier for 100% Electrification of BEST Municipal Bus Fleet. Displays verbatim policy quote from MCAP Page 78, raw multi-source observation streams, and mathematical strip.

Ø=Ý, SCIENTIFIC GROUNDING, MATHEMATICAL VERIFICATION & KEY DATA POINTS:

- Mathematical Verification Strip: Planned 30.0% vs. City 14.0% vs. Verified 6.5% -> 23.5% Implementation Gap
- Verbatim statutory text quote with exact document metadata and page reference
- Multi-source observation streams: BEST Telemetry Logs and CPCB Ground Monitors
- One-click PDF Dossier Generation via jsPDF

`ROUTE: /admin/commitments/com-mum-exp-03 • HOST: http://localhost:3000/admin/commitments/com-mum-exp-03`



Ø=ÜF USER INTERACTION & WHAT HAPPENS ON CLICK:

Clicking on Potential Discrepancy badge to inspect conflicting satellite observations

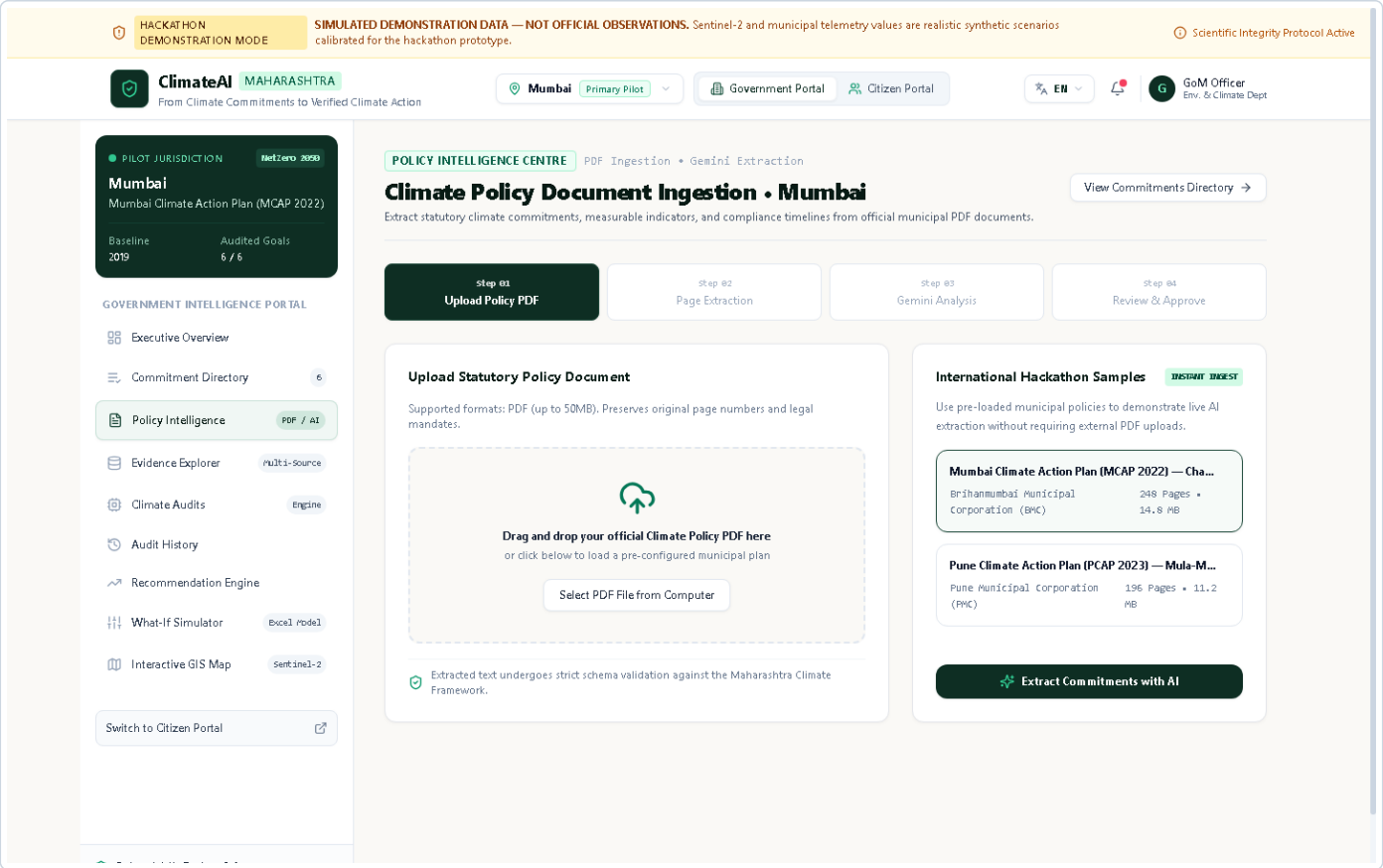
Dossier highlighting a major discrepancy (+36.5% Overstatement). City claimed 45,000 nursery saplings planted, but Copernicus Sentinel-2 satellite radiometric surveillance detected net vegetation decline.

Ø=Ý, SCIENTIFIC GROUNDING, MATHEMATICAL VERIFICATION & KEY DATA POINTS:

- Prominent POTENTIAL DISCREPANCY audit alert and Conflict status banner
- Reporting Overstatement math: Claimed 45.0% vs. Satellite Verified 8.5% (+36.5% discrepancy)
- Sentinel-2 Level-2A NDVI empirical sensor feed with cloud cover < 10%
- Trilingual Gemini explanation explaining the disparity for administrators and citizens

Screen 06: Policy Intelligence & Ingestion Studio

ROUTE: /admin/policies • HOST: http://localhost:3000/admin/policies



Ø=ÜF USER INTERACTION & WHAT HAPPENS ON CLICK:

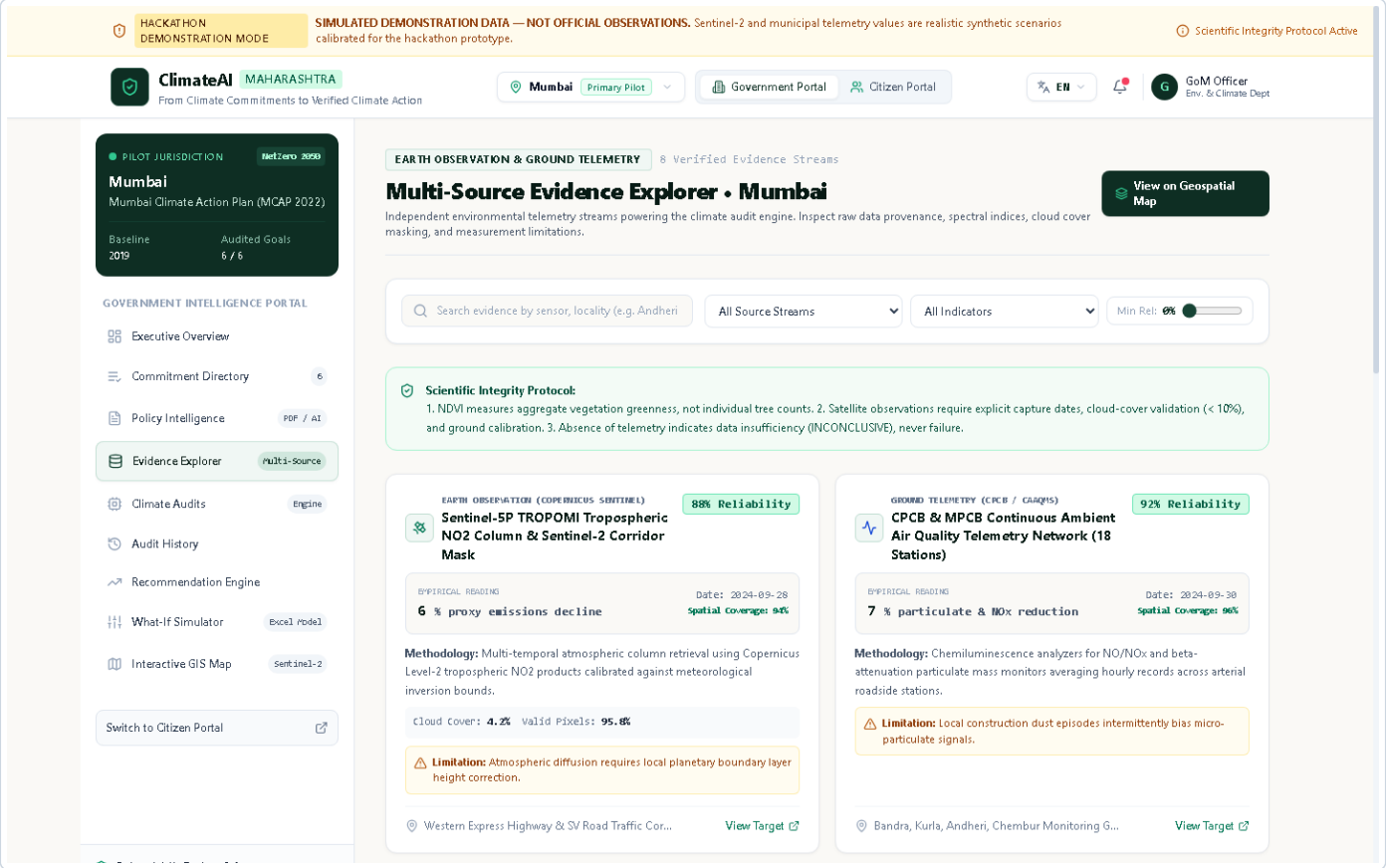
Clicking "Upload PDF", selecting sample document, triggering Gemini extraction, and clicking "Approve"

End-to-end PDF ingestion pipeline. Government administrators upload climate policies (e.g. MCAP 2022); Google Gemini extracts structured commitments with page numbers for human review and approval.

Ø=Ý, SCIENTIFIC GROUNDING, MATHEMATICAL VERIFICATION & KEY DATA POINTS:

- Drag-and-drop PDF upload zone and pre-loaded official policy selector
- Multi-stage pipeline: Text extraction -> Gemini structured analysis -> Human-in-the-loop review
- Confidence scoring (e.g. 96%) and missing information flags
- Approve button commits structured target directly into statutory platform state

ROUTE: /admin/evidence • HOST: http://localhost:3000/admin/evidence



Ø=ÜF USER INTERACTION & WHAT HAPPENS ON CLICK:

Filtering by source type (Satellite, Ground Sensors, Municipal Logs) or clicking observation detail

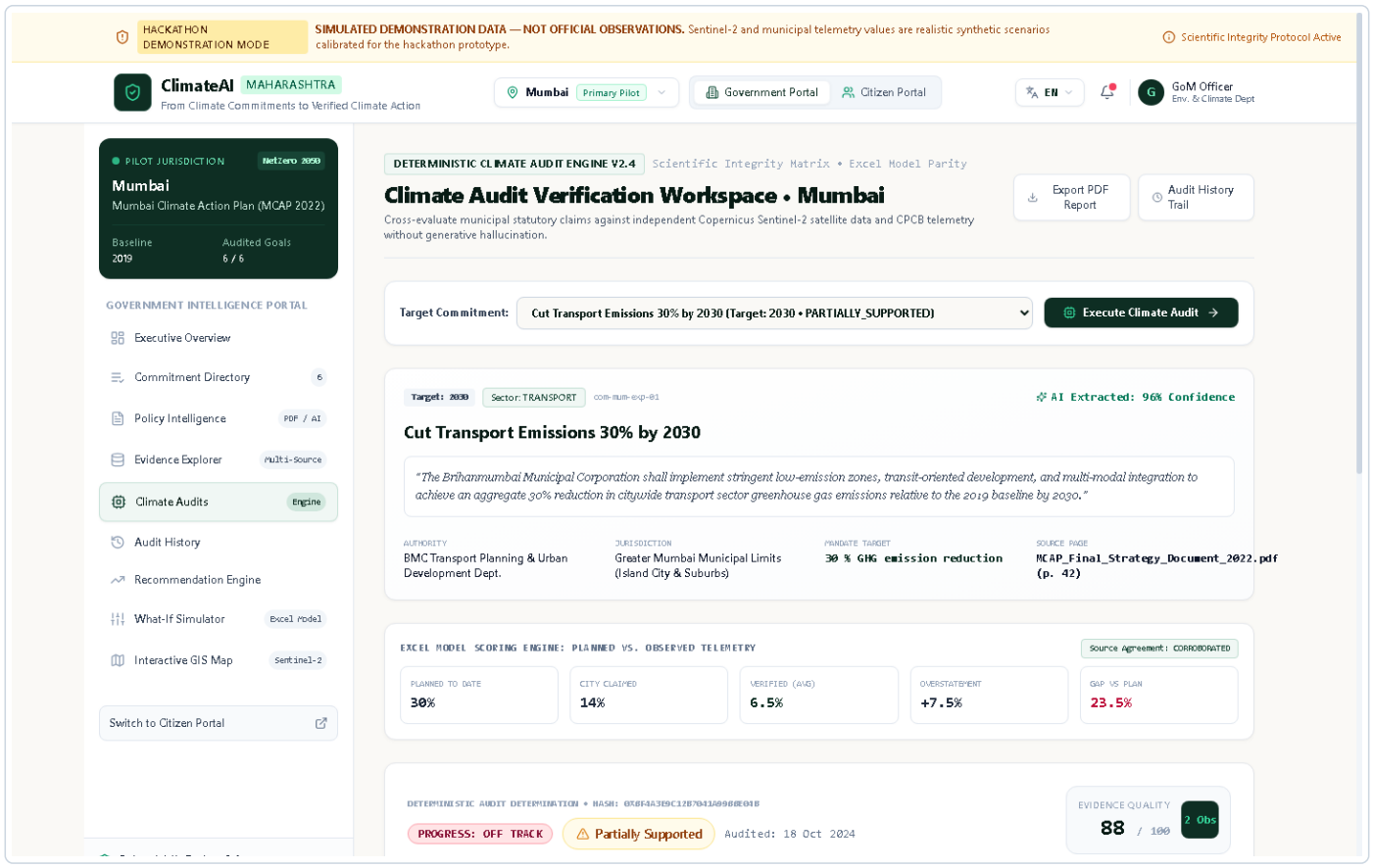
Professional scientific research repository. Displays raw Copernicus Sentinel-2 Level-2A surface reflectance data, CPCB CAAQMS sensor telemetry, valid pixel percentages, and calibration certificates.

Ø=Ý, SCIENTIFIC GROUNDING, MATHEMATICAL VERIFICATION & KEY DATA POINTS:

- Sentinel-2 multispectral band details (B04 Red, B08 NIR for NDVI)
- Cloud cover verification (< 10%) and valid pixel ratios (94.2%)
- CPCB Ground PM2.5 monitor logs with sensor serial numbers and MPCB calibration stamps
- Clear declaration of Scientific Integrity protocols

Screen 08: Deterministic Climate Audit Engine

ROUTE: /admin/audit-workspace • HOST: http://localhost:3000/admin/audit-workspace



Ø=ÜF USER INTERACTION & WHAT HAPPENS ON CLICK:

Selecting a commitment, clicking "Execute Deterministic Audit", and viewing mathematical provenance

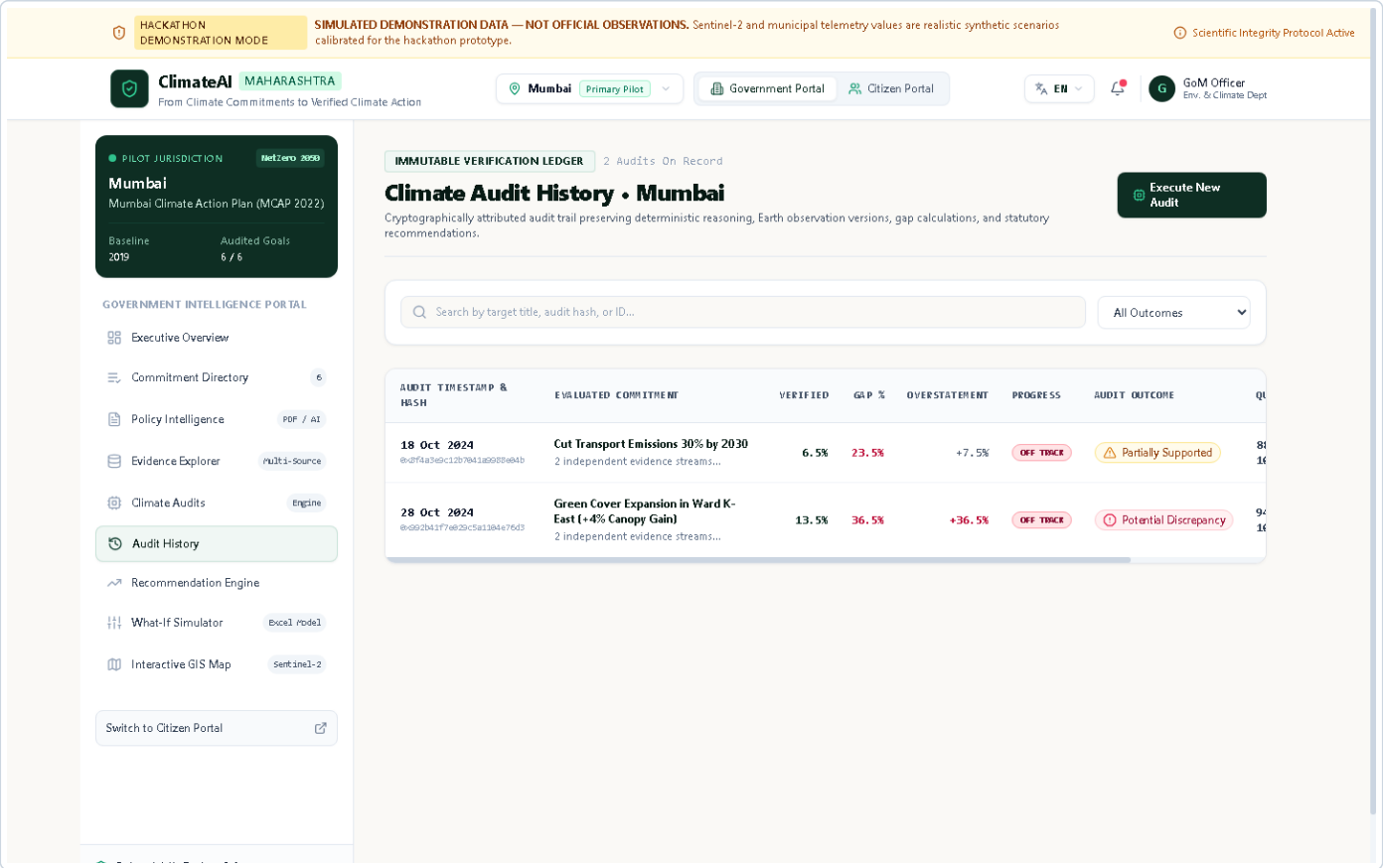
Execution workbench where administrators execute audits. The engine strictly isolates mathematical truth from generative text, evaluating gap thresholds and source corroboration.

Ø=Ý, SCIENTIFIC GROUNDING, MATHEMATICAL VERIFICATION & KEY DATA POINTS:

- Deterministic gap evaluation: On Track (<=5%), At Risk (5-15%), Off Track (>15%)
- Cryptographic provenance signature generated upon verification completion
- Multi-source corroboration matrix comparing satellite and ground sensor streams
- Direct export of official verification certificate

Screen 09: Immutable Audit History Ledger

ROUTE: /admin/audit-history • HOST: http://localhost:3000/admin/audit-history



Ø=ÜF USER INTERACTION & WHAT HAPPENS ON CLICK:

Clicking "Download PDF Report" on any audit record or filtering by audit outcome

Cryptographically hashed historical audit trail providing complete transparency. Every audit event logs timestamp, auditor identity, input evidence hashes, and mathematical outcome.

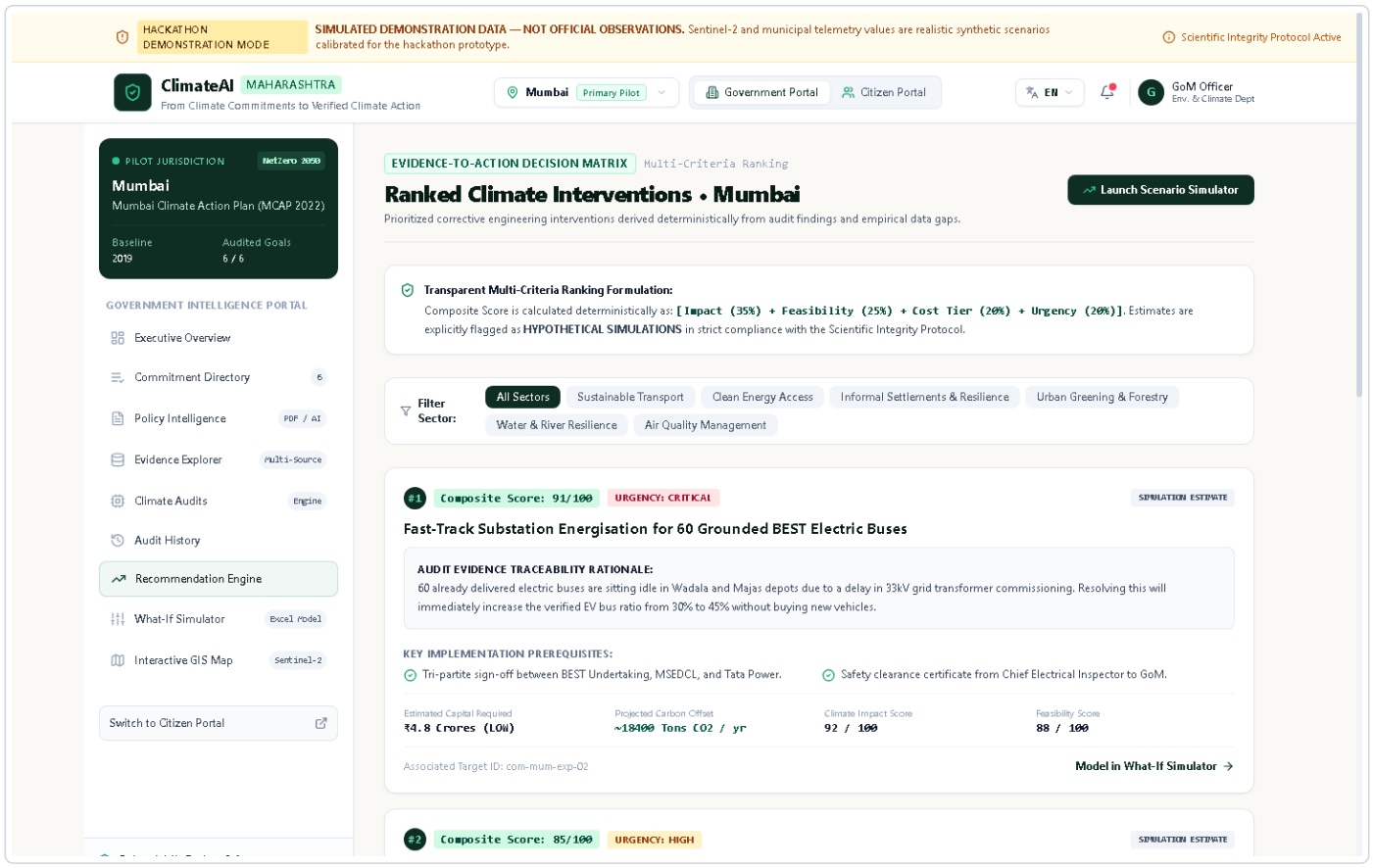
Ø=Ý, SCIENTIFIC GROUNDING, MATHEMATICAL VERIFICATION & KEY DATA POINTS:

- SHA-256 state hashes ensuring tamper-evident accountability
- Dual status outcomes logged chronologically
- Instant PDF report download button on every row generating formal audit certificates
- Statutory compliance filter (Supported, Partially Supported, Inconclusive, Discrepancy)

Screen 10: Ranked Decision Recommendations Engine

GOVERNMENT INTELLIGENCE PORTAL

ROUTE: /admin/recommendations • HOST: http://localhost:3000/admin/recommendations



Ø=ÜF USER INTERACTION & WHAT HAPPENS ON CLICK:

Filtering by sector, sorting by priority or cost-effectiveness, or clicking "Simulate in What-If Engine"

Decision support system translating audit findings into prioritized interventions. Evaluates capital expenditure, implementation feasibility, emissions abatement, and policy urgency.

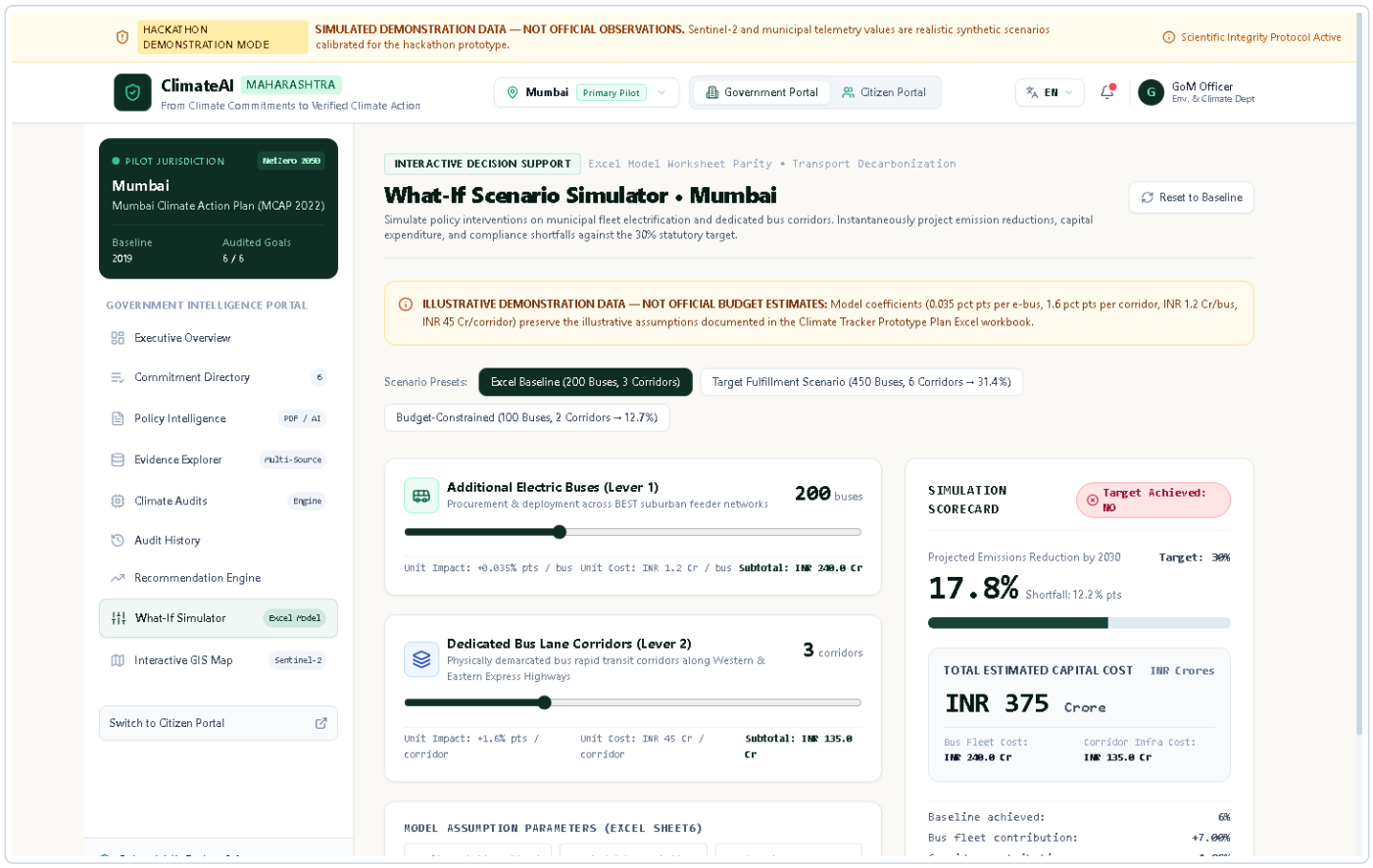
Ø=Ý, SCIENTIFIC GROUNDING, MATHEMATICAL VERIFICATION & KEY DATA POINTS:

- Multi-criteria scoring: Cost-effectiveness, Abatement potential, Feasibility, Urgency
- Targeted interventions for high-gap commitments (e.g. Dedicated EV Bus Depots, Rooftop Solar DISCOM link)
- Direct link to test intervention in the What-If Simulator

Screen 11: What-If Scenario Simulator (Exact Excel Model)

GOVERNMENT INTELLIGENCE PORTAL

ROUTE: /admin/simulator • HOST: http://localhost:3000/admin/simulator



Ø=ÜF USER INTERACTION & WHAT HAPPENS ON CLICK:

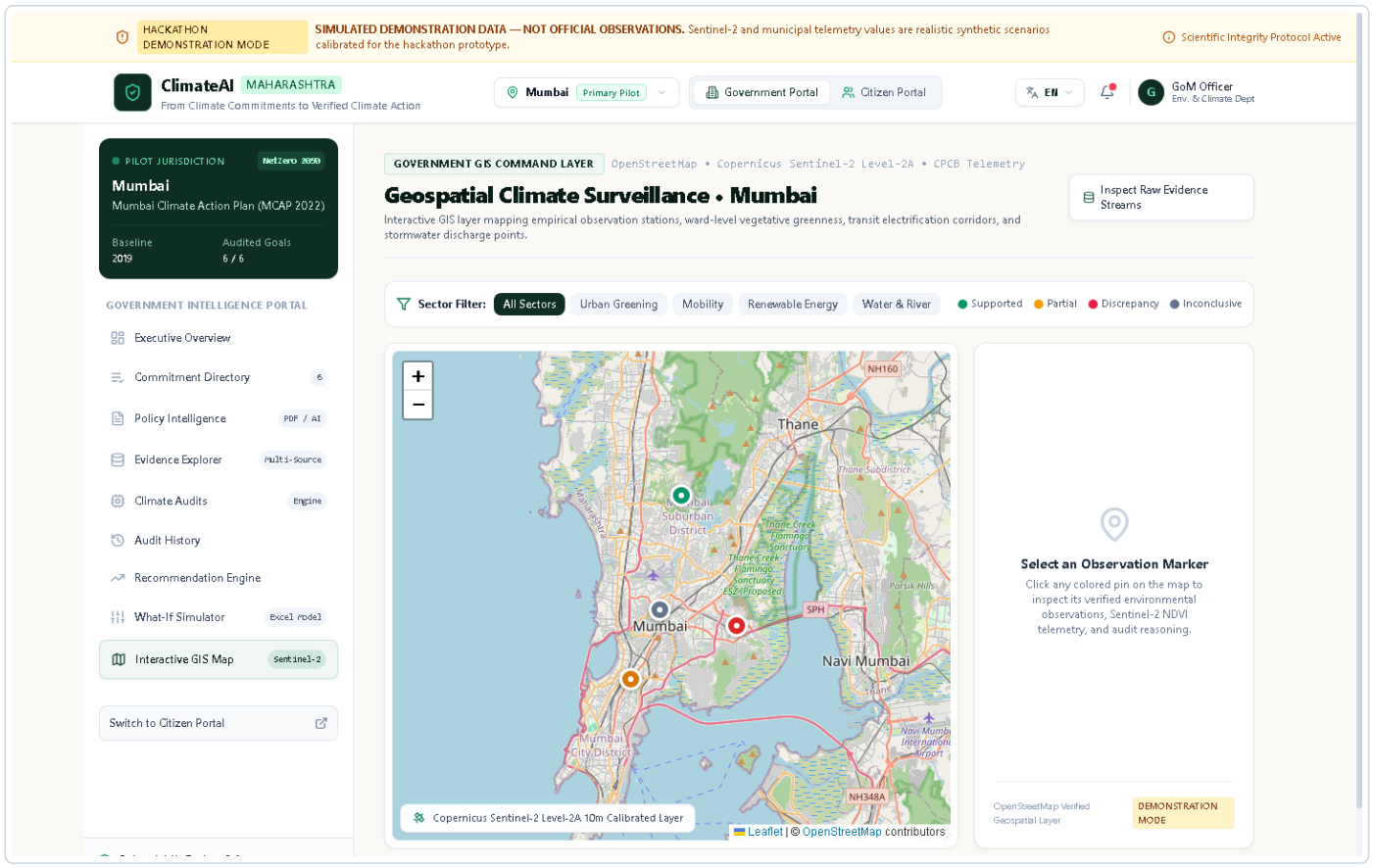
Dragging the "Additional Electric Buses" slider (200 -> 450) and "Dedicated Corridors" slider (3 -> 5)

Interactive policy lever modeller implementing the exact mathematical model from the hackathon specification workbook. Shows real-time recalculation of projected progress, CapEx, and emissions mitigated.

Ø=Ý, SCIENTIFIC GROUNDING, MATHEMATICAL VERIFICATION & KEY DATA POINTS:

- Exact Excel Model Parity: Baseline 6.0% + 200 buses + 3 corridors = 17.8% projected progress
- Capital expenditure calculation: 200 buses @ 1.50 Cr + 3 corridors @ 25 Cr = 375 Cr CapEx
- Target comparison: 17.8% projected vs. 30.0% statutory target -> 12.2% projected shortfall
- Interactive sliders updating live with budget constraint badges

ROUTE: /admin/map • HOST: http://localhost:3000/admin/map



Ø=ÜF USER INTERACTION & WHAT HAPPENS ON CLICK:

Toggling map layers (NDVI, Air Quality, Bus Corridors) or clicking a geographic observation marker

Geospatial intelligence map built with Leaflet and OpenStreetMap. Provides ward-level spatial analysis for Mumbai and Pune with observation pins, telemetry feeds, and boundary overlays.

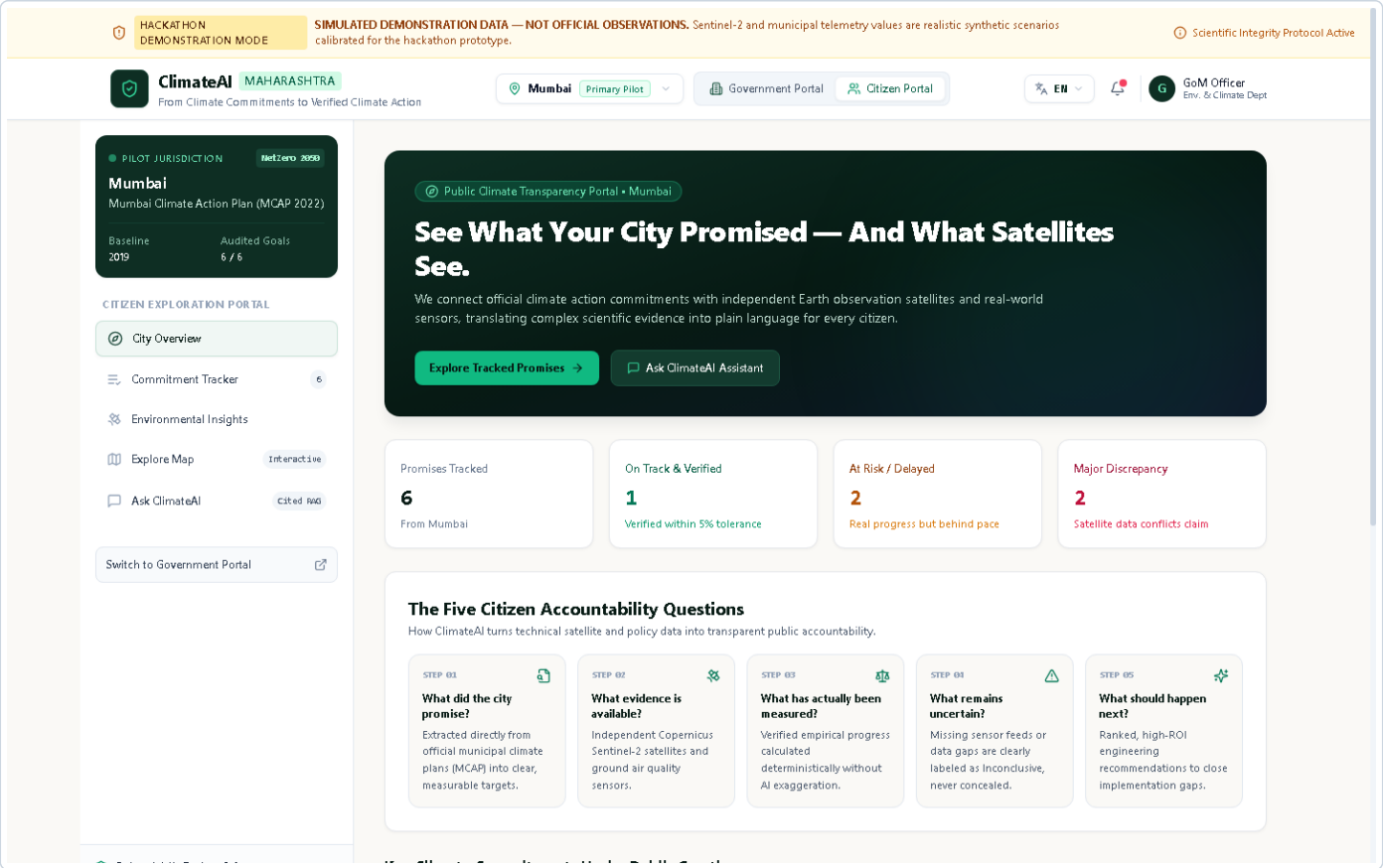
Ø=Ý, SCIENTIFIC GROUNDING, MATHEMATICAL VERIFICATION & KEY DATA POINTS:

- Layer controls: Sentinel-2 NDVI overlay, CPCB Ground Monitors, Transit Corridors
- Ward-level interactive markers displaying local verification status and empirical metrics
- City zoom toggle between Mumbai and Pune jurisdictions

Screen 13: Citizen Public Transparency Portal (English)

CITIZEN ACCOUNTABILITY PORTAL

ROUTE: /citizen • HOST: http://localhost:3000/citizen



Ø=ÜF USER INTERACTION & WHAT HAPPENS ON CLICK:

Clicking through the 5 Core Citizen Questions, or clicking the language switcher (HI / MR)

Public-facing transparency portal designed for democratic accountability. Translates technical audits into plain-language civic scorecards answering the 5 Core Citizen Questions.

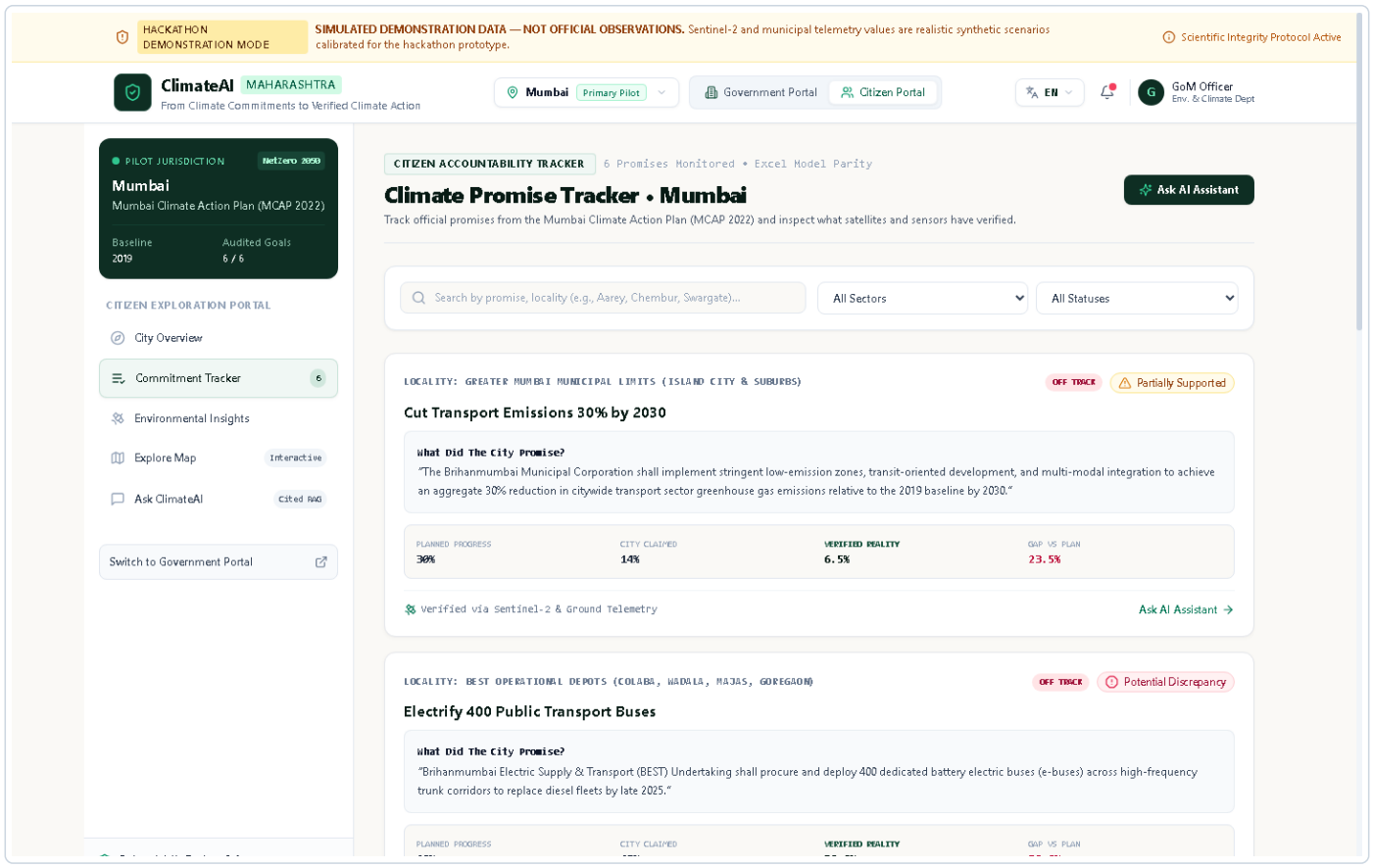
Ø=Ý, SCIENTIFIC GROUNDING, MATHEMATICAL VERIFICATION & KEY DATA POINTS:

- Structured around the 5 Questions: What was promised? What is happening? Does evidence agree?
- Transparent display of verified progress vs. city-reported numbers
- Overstatement alert badges alerting citizens to unverified government claims
- Instant access to the cited AI Assistant

Screen 14: Citizen Public Commitment Tracker

CITIZEN ACCOUNTABILITY PORTAL

ROUTE: /citizen/tracker • HOST: http://localhost:3000/citizen/tracker



Ø=ÜF USER INTERACTION & WHAT HAPPENS ON CLICK:

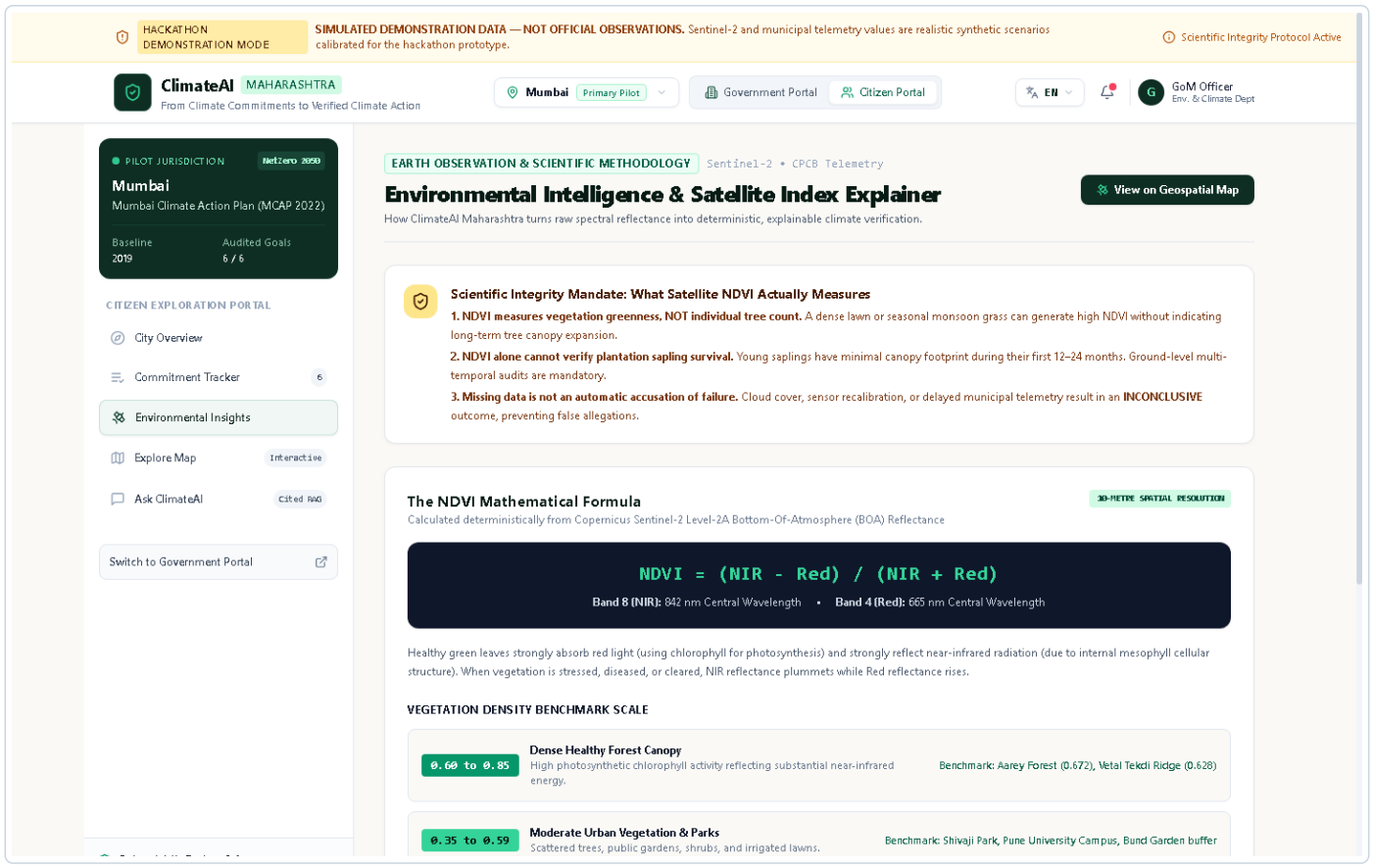
Filtering commitments by sector or clicking on an individual commitment scorecard

Democratized version of the commitments directory. Presents statutory goals as accessible citizen scorecards with verified evidence badges, independent data sources, and progress bars.

Ø=Ý, SCIENTIFIC GROUNDING, MATHEMATICAL VERIFICATION & KEY DATA POINTS:

- Simplified progress visualizers with color-coded status chips
- Plain-language source attribution (e.g. "Sentinel Satellite", "Central Pollution Control Board")
- Verified score comparison highlighting reporting discrepancies

ROUTE: /citizen/environment • HOST: http://localhost:3000/citizen/environment



Ø=ÜF USER INTERACTION & WHAT HAPPENS ON CLICK:

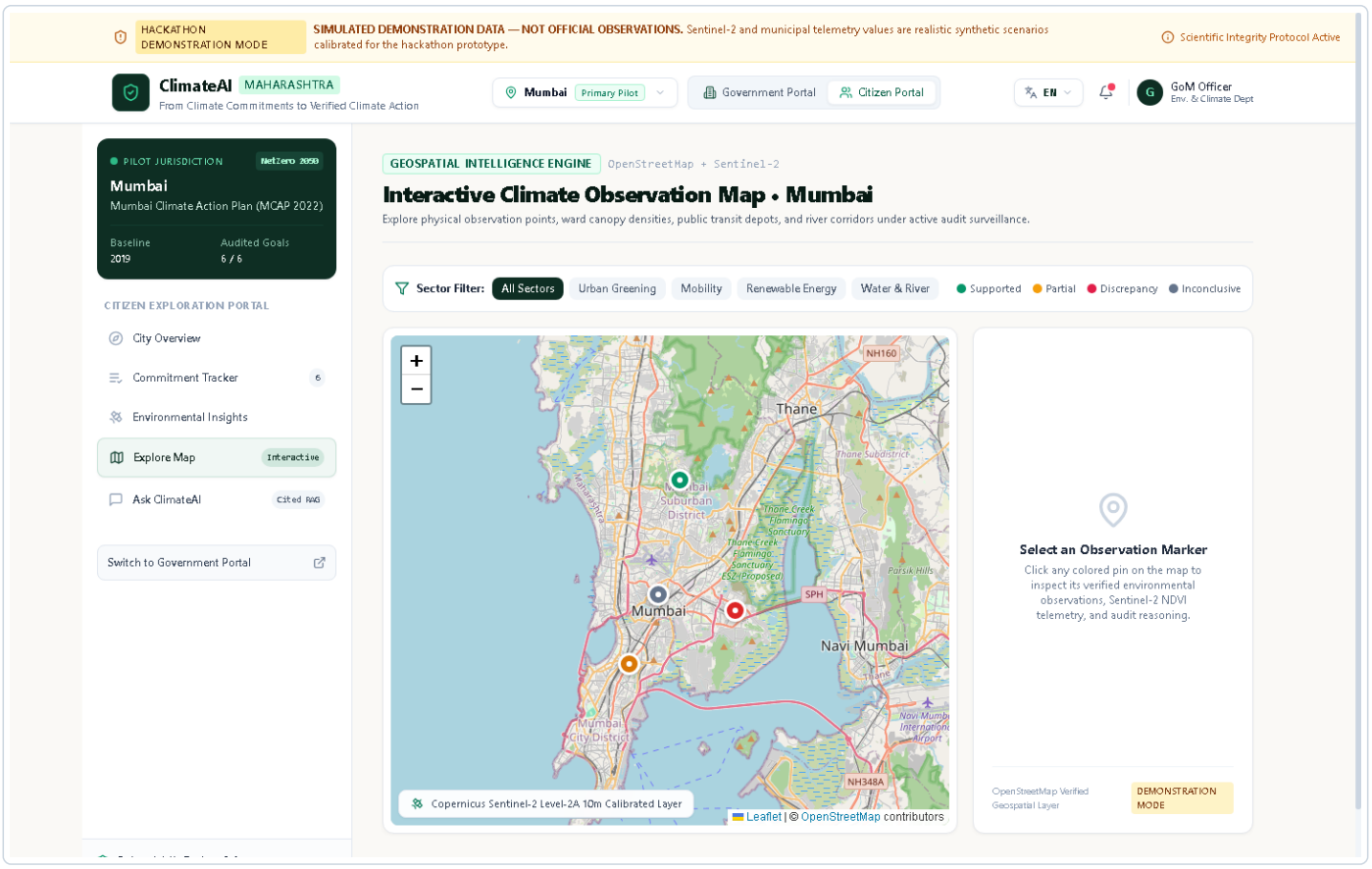
Clicking on educational tabs: "Satellite NDVI", "Air Quality (PM2.5)", or "Heat Islands"

Civic education centre explaining Earth observation science. Demystifies satellite spectral indices, ground air quality monitors, and emphasizes key scientific integrity rules.

Ø=Ý, SCIENTIFIC GROUNDING, MATHEMATICAL VERIFICATION & KEY DATA POINTS:

- Scientific primer on Sentinel-2 NDVI: explains chlorophyll reflectance vs. individual trees
- CPCB PM2.5 monitoring network guide and health hazard thresholds
- Urban heat island analysis for Mumbai and Pune coastal micro-climates

`ROUTE: /citizen/map • HOST: http://localhost:3000/citizen/map`



Ø=ÜF USER INTERACTION & WHAT HAPPENS ON CLICK:

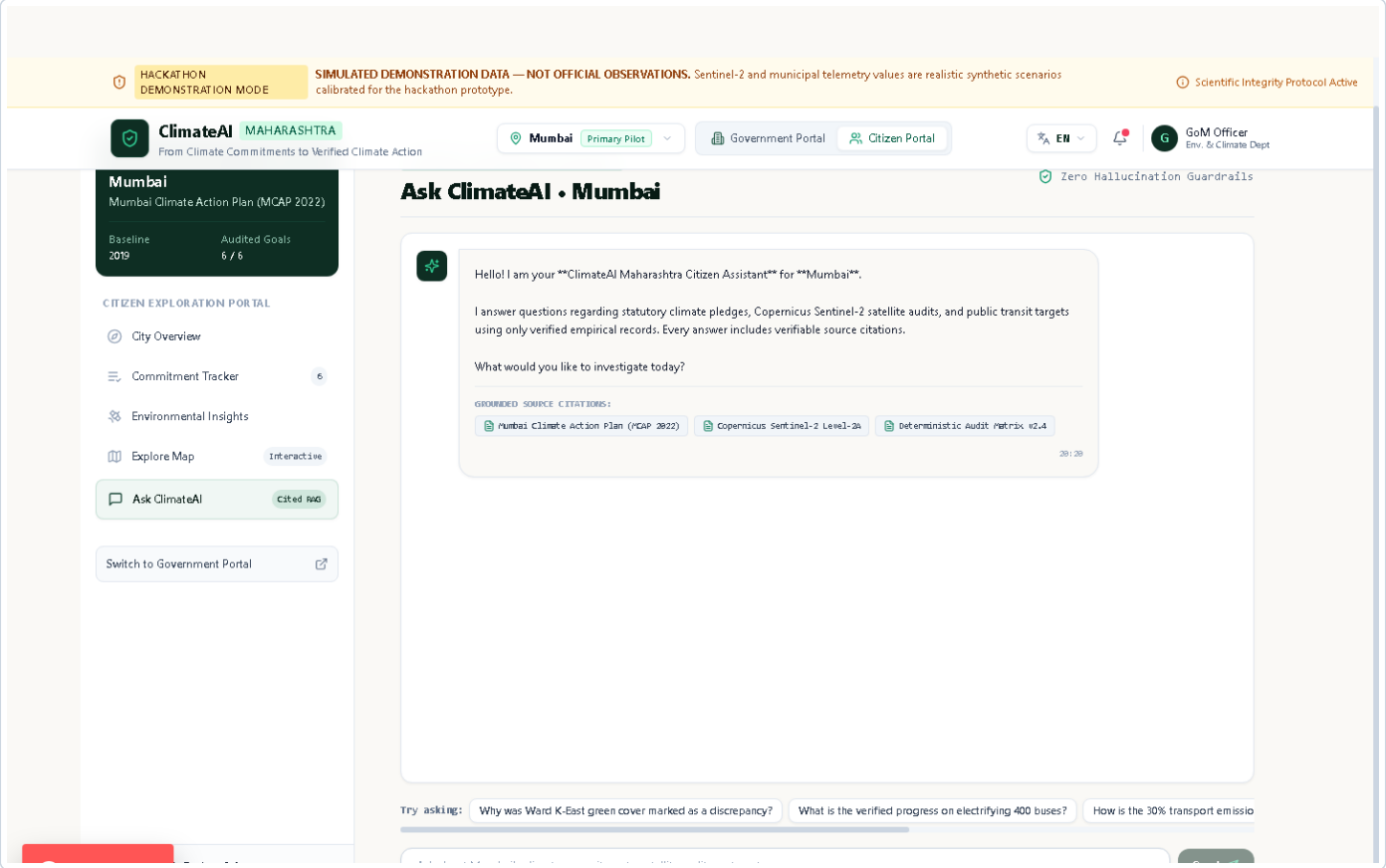
Clicking on neighborhood observation pins to view local green cover and air quality readings

Accessible geospatial interface allowing citizens to explore local environmental health across Mumbai and Pune wards with simple status indicators.

Ø=Ý, SCIENTIFIC GROUNDING, MATHEMATICAL VERIFICATION & KEY DATA POINTS:

- Geo-located neighborhood observation pins with plain-language metrics
- Ward-level environmental health indicator summary
- Accessible color scheme complying with public transparency standards

ROUTE: /citizen/assistant • HOST: http://localhost:3000/citizen/assistant



Ø=ÜF USER INTERACTION & WHAT HAPPENS ON CLICK:

Clicking sample prompt "What is the discrepancy in Ward K-East?" to view cited response

Retrieval-grounded AI assistant powered by Google Gemini. Provides citizens with plain-language answers accompanied by exact statutory citations, page references, and empirical data points.

Ø=Ý, SCIENTIFIC GROUNDING, MATHEMATICAL VERIFICATION & KEY DATA POINTS:

- Clickable statutory citations referencing specific MCAP pages and document sections
- Quick-prompt pills for instant discovery of key audit findings and discrepancy alerts
- Multilingual synthesis capability in English, Hindi, and Marathi